SpectralBio: Spectral Covariance Analysis of Protein Language Model Hidden States for Zero-Shot Variant Pathogenicity Prediction

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Abstract. Predicting whether missense mutations are pathogenic or benign from sequence alone is a central challenge in computational biology. State-of-the-art zero-shot methods leverage protein language models (PLMs) via log-likelihood ratios, but these capture only sequence probability and ignore the rich geometry of internal representations. We present SpectralBio, a framework that analyzes the *spectral and geometric* structure of ESM2-150M hidden-state covariance matrices to derive pathogenicity signals complementary to likelihood scores. For each variant we compute local covariance matrices (±40-residue window) and extract three features: Frobenius covariance distance (*FrobDist*), trace ratio (*TraceRatio*), and log-eigenvalue shift (*SPS-log*). On ClinVar TP53 (N=255; 115 pathogenic, 140 benign), **FrobDist + LL Proper** achieves **AUC = 0.7498** on the TP53 canonical executable benchmark. On the secondary bounded transfer evaluation without retraining, proper masked-LM LL reaches **AUC = 0.9174** on a fixed BRCA1 subset (N=100), providing secondary bounded transfer evidence only. Matrix-level features consistently outperform eigenvalue-only SPS on TP53, and the TP53 canonical benchmark reproduces with $\Delta_{\mathbf{rep}}$**=0.0**.

1 Introduction

Identifying disease-causing protein variants from sequence data alone remains one of the most pressing challenges in computational biology. As whole-genome sequencing becomes routine, the number of variants of uncertain significance (VUS) grows far faster than experimental capacity to characterize them [3]. Computational predictors that can reliably rank variants without labeled training data are therefore useful research tools for zero-shot variant effect prediction.

Protein language models (PLMs), particularly ESM-2 [1], have emerged as powerful zero-shot predictors. By computing log-likelihood ratios (ΔLLR) between wild-type and mutant sequences [2], PLMs achieve competitive accuracy without supervised fine-tuning. However, standard LLR implementations capture only the probability of each sequence under the model's masked-language-modeling objective, discarding information encoded in the *geometry of internal representations*.

Recent work in adversarial ML showed that spectral analysis of hidden-state covariance can detect subtle perturbations [4]. We hypothesize that disease-causing mutations induce detectable distortions in PLM hidden-state covariance, providing signal *orthogonal* to likelihood-based scores.

Contributions. **(1)** We introduce SpectralBio, extracting matrix-level spectral features from PLM hidden-state covariance. **(2)** On the TP53 canonical executable benchmark, FrobDist + proper LL yields AUC=0.7498. **(3)** On a secondary bounded transfer evaluation without retraining, Proper LL achieves AUC=0.9174 on a fixed BRCA1 subset (N=100). **(4)** We compare matrix-level vs. eigenvalue-only spectral features for variant prediction within this artifact. **(5)** The TP53 canonical benchmark reproduces with Δ_{rep}=0.0 under the repository's canonical execution path.

2 Related Work

Zero-shot variant effect prediction. Meier et al. [2] showed that ESM-1b log-likelihood ratios achieve strong zero-shot variant effect prediction. Notin et al. [5] established ProteinGym benchmarks spanning hundreds of proteins, confirming that larger PLMs generally improve AUC with diminishing returns beyond 650M parameters. All these methods operate purely on sequence probability.

Spectral analysis of neural representations. Bonetto [4] introduced SpectralGuard, which monitors the spectral radius of hidden-state covariance in Mamba-based SSMs to detect poisoning attacks—demonstrating that input perturbations produce detectable eigenspectrum shifts. SpectralBio applies this insight to PLM hidden states, showing that geometric features complement likelihood scores for pathogenicity prediction.

3 Method

3.1 Model and Dataset

We use ESM2-150M (`esm2_t30_150M_UR50D`; 150M parameters, L=30 layers, d=640) [1] with its masked-LM head. The model is used entirely in inference mode with no fine-

tuning. Missense variants are sourced from ClinVar [3], filtered for "Pathogenic" and "Benign" labels. Primary evaluation: TP53 (N=255; 115 pathogenic, 140 benign). Secondary bounded transfer evaluation: bounded transfer on a fixed BRCA1 subset (N=100) without retraining. Operationally, this artifact is anchored to the *TP53 canonical executable benchmark*; the BRCA1 subset is a *secondary transfer evaluation without retraining*, and any broader reuse is treated as an *adaptation recipe only*.

3.2 Spectral Feature Extraction

Given wild-type $\mathbf{S}_{\mathrm{WT}}$ and mutant $\mathbf{S}_{\mathrm{MUT}}$ at position p, extraction proceeds in four steps:

Step 1: Local windowing. We extract ±40 residues around the mutation site (subsequence length $w \leq 81$), focusing analysis on the structural neighborhood most likely perturbed.

Step 2: Hidden state extraction. Both windows are passed through ESM2-150M. For each layer $\ell \in \{1, \ldots, L\}$, we extract hidden states $\mathbf{H}^{(\ell)} \in \mathbb{R}^{w \times d}$.

Step 3: Covariance. Residue-level covariance $\mathbf{C}^{(\ell)} = \mathrm{Cov}(\mathbf{H}^{(\ell)}) \in \mathbb{R}^{d \times d}$ is computed separately for wild-type and mutant.

Step 4: Spectral features. Three features averaged over all L=30 layers:

$$\mathrm{FrobDist} = \frac{1}{L} \sum_{\ell=1}^{L} \left\| \mathbf{C}_{\mathrm{MUT}}^{(\ell)} - \mathbf{C}_{\mathrm{WT}}^{(\ell)} \right\|_F \tag{1}$$

$$\mathrm{TraceRatio} = \frac{1}{L} \sum_{\ell=1}^{L} \left| \frac{\mathrm{tr}(\mathbf{C}_{\mathrm{MUT}}^{(\ell)})}{\mathrm{tr}(\mathbf{C}_{\mathrm{WT}}^{(\ell)})} - 1 \right| \tag{2}$$

$$\text{SPS-log} = \frac{1}{L} \sum_{\ell=1}^{L} \left\| \log |\boldsymbol{\lambda}_{\mathrm{MUT}}^{(\ell)}| - \log |\boldsymbol{\lambda}_{\mathrm{WT}}^{(\ell)}| \right\|_2^2 \tag{3}$$

where $\boldsymbol{\lambda}^{(\ell)}$ denotes the eigenvalues of $\mathbf{C}^{(\ell)}$. We treat Eqs. 1–3 as the core spectral equation set of SPECTRAL-BIO: full-matrix displacement, trace-scale perturbation, and eigenvalue-log shift, respectively. FrobDist (Eq. 1) captures total covariance perturbation magnitude. TraceRatio (Eq. 2) measures variance changes. SPS-log (Eq. 3) uses only eigenvalues, discarding eigenvector information.

3.3 Log-Likelihood Scores

(a) LL Crude—hidden-state norm difference proxy (baseline, AUC=0.7026). **(b) LL Proper**—direct masked-LM log-likelihood:

$$\mathrm{LL}(v) = \log P_{\mathrm{ESM2}}(r_p^{\mathrm{wt}} | \mathbf{S}_{\mathrm{WT}}) - \log P_{\mathrm{ESM2}}(r_p^{\mathrm{mut}} | \mathbf{S}_{\mathrm{WT}}) \tag{4}$$

A positive $\mathrm{LL}(v)$ indicates the model considers the wild-type residue more likely, suggesting the mutation is deleterious.

3.4 Evaluation Protocol

Features are normalized to $[0, 1]$ via MinMaxScaler. Pairwise combinations: grid search over $\alpha \in [0, 1]$ (step 0.05) with $s = \alpha \cdot f_1 + (1 - \alpha) \cdot f_2$. AUC-ROC with 95% bootstrap CIs (B=1,000). All seeds fixed to 42.

Table 1: **TP53 pathogenicity classification (AUC-ROC).** N=255: 115 pathogenic, 140 benign. Δ_{rep}=0.0.

Method	**AUC**
Individual features	
LL Crude (norm proxy, baseline)	0.7026
TraceRatio (spectral, matrix)	0.6242
FrobDist (spectral, matrix)	0.6209
SPS-log (spectral, eigenvalue)	0.5988
LL Proper (masked LM)	0.5956
SPS-all (spectral, eigenvalue)	0.5611
SPS-topk (spectral, eigenvalue)	0.5606
SPS-deep10 (spectral, eigenvalue)	0.5542
SPS-deep5 (spectral, eigenvalue)	0.5298
Best combinations	
FrobDist + LL Proper (α=0.55)	**0.7498**
LL Crude + Trace + Frob	0.7264

Table 2: **Secondary bounded transfer evaluation on fixed BRCA1 subset (AUC-ROC).** N=100. Official reported metric only. No retraining or recalibration.

Feature	**AUC**
LL Proper (masked LM)	**0.9174**

4 Results

4.1 TP53 Classification

Table 1 presents the full TP53 results. Among individual features, LL Crude leads (0.7026), followed by the matrix-level spectral features TraceRatio (0.6242) and FrobDist (0.6209). Eigenvalue-only SPS variants perform substantially worse (SPS-log: 0.5988; SPS-all: 0.5611; SPS-topk: 0.5606; SPS-deep10: 0.5542; SPS-deep5: 0.5298), confirming that retaining full covariance structure is more informative than eigenvalue summaries alone.

The best combination, **FrobDist + LL Proper** (α=0.55/0.45), achieves **AUC=0.7498** on the TP53 canonical executable benchmark. Relative to LL Crude, this corresponds to a descriptive within-benchmark gain of +0.0472 AUC (+6.7% relative). The best triple (LL Crude + Trace + Frob; 0.70/0.10/0.20) yields AUC=0.7264—lower than the best pair, suggesting crude LL and proper LL capture partially overlapping information that interferes when combined.

4.2 Secondary Bounded Transfer Evaluation (BRCA1)

Table 2 presents the secondary bounded transfer results on the fixed BRCA1 subset. Proper LL achieves **AUC=0.9174** (N=100) on this fixed subset without retraining. This secondary bounded transfer evaluation is included as fixed-subset evidence only and does not establish broader cross-protein behavior.

Within this bounded setting, the fixed BRCA1 subset result shows that the proper masked-LM score can remain informative without retraining. We report this result as secondary transfer evidence only and treat any broader reuse beyond TP53 and this fixed subset as an adaptation

recipe only.

4.3 Matrix-Level vs. Eigenvalue-Only Features

Within the TP53 results reported here, matrix-level spectral features (FrobDist, TraceRatio) outperform eigenvalue-only summaries (SPS variants). On TP53, both FrobDist (0.6209) and TraceRatio (0.6242) exceed all SPS variants (best: 0.5988). This pattern suggests that retaining full covariance structure may preserve useful benchmark signal beyond eigenvalue-only summaries.

4.4 Visual Analysis

Figure 1 provides a comprehensive visual summary. The ROC plot (top-left) shows the Best Pair combination (orange, FrobDist+LL Proper, AUC=0.7498) remaining above all individual features and the Best Triple (black, 0.7264, rounded in the figure), with SPS variants (dashed) clustered near the diagonal. The feature ablation bar chart (top-right) ranks all individual AUCs, with LL Crude (red, 0.7026, rounded in the figure) as the strongest standalone feature and SPS/positional entropy (gray) at the bottom—visually confirming the matrix > eigenvalue hierarchy. The SPS-log density plot (bottom-left) reveals why eigenvalue-only features underperform: both pathogenic (red) and benign (blue) distributions are heavily concentrated near zero, with minimal class separation. In contrast, the proper LL distribution (bottom-right) shows a clear rightward shift for pathogenic variants (higher $\log P(\text{wt}) - \log P(\text{mut})$), consistent with the secondary bounded BRCA1 transfer evaluation.

5 Discussion

SpectralBio demonstrates that covariance geometry of PLM hidden states can provide useful benchmark signal complementary to sequence-level log-likelihoods.

Why feature complementarity works. The best TP53 result combines a spectral feature (FrobDist, geometric perturbation) with a likelihood feature (LL Proper, evolutionary constraint). These are mechanistically distinct: FrobDist measures how much the *shape* of the representation manifold changes, while LL Proper measures how *surprising* the mutant residue is. Their complementarity suggests pathogenic mutations affect both the surface-level amino acid preference and the deeper structural representation independently.

Why LL Proper matters in the bounded BRCA1 transfer evaluation. Proper LL achieving AUC=0.9174 on the fixed BRCA1 subset despite 0.5956 on TP53 shows that this score remains informative in the secondary fixed-subset setting. We keep the interpretation narrow: this bounded result does not establish broader cross-protein behavior beyond the fixed BRCA1 subset evaluated here.

Key findings:

1. **Matrix > eigenvalue.** FrobDist/TraceRatio consistently outperform all SPS variants on TP53.
2. **Feature complementarity.** Strongest TP53 classifier (0.7498) combines spectral geometry with masked-LM likelihood—neither is strongest alone.
3. **Secondary BRCA1 bounded transfer evidence.** LL Proper: 0.9174 with zero retraining on the fixed subset.
4. **TP53 canonical reproducibility.** Δ_{rep}=0.0 under the repository's canonical execution path.

Limitations. Window size (±40), model size (150M), and mixing weights were not jointly optimized. SPS features remain weak (best: 0.5988); layer selection provided no improvement. Evaluation limited to two proteins; ProteinGym [5] benchmarking with 87 proteins is future work. Larger ESM2 variants (650M–15B) likely produce richer geometry.

6 Reproducibility

Seeds: 42 (PyTorch, NumPy, `random`). This repository is distributed as a research reproducibility artifact anchored to the TP53 canonical executable benchmark and a secondary bounded BRCA1 transfer evaluation without retraining. Model: ESM2-150M (HuggingFace, public). Dataset: ClinVar FTP (public). Δ_{rep}=0.0 for the TP53 canonical executable benchmark. Code: github.com/DaviBonetto/SpectralBio. Demo: huggingface.co/spaces/DaviBonetto/spectralbio-demo.

Dataset: huggingface.co/datasets/DaviBonetto/spectralbio-clinvar.

7 Conclusion

We presented SpectralBio, a zero-shot variant pathogenicity prediction framework based on spectral analysis of PLM hidden-state covariance. Matrix-level spectral features provide complementary signal to log-likelihood ratios, yielding AUC=0.7498 on TP53 and AUC=0.9174 on a secondary bounded transfer evaluation on a fixed BRCA1 subset (N=100) without retraining. The consistent superiority of covariance-based features over eigenvalue-only summaries suggests that full covariance structure may capture useful benchmark signal underexplored by eigenvalue-only summaries. These results are presented as a research reproducibility artifact with one primary benchmark and one bounded secondary transfer evaluation.

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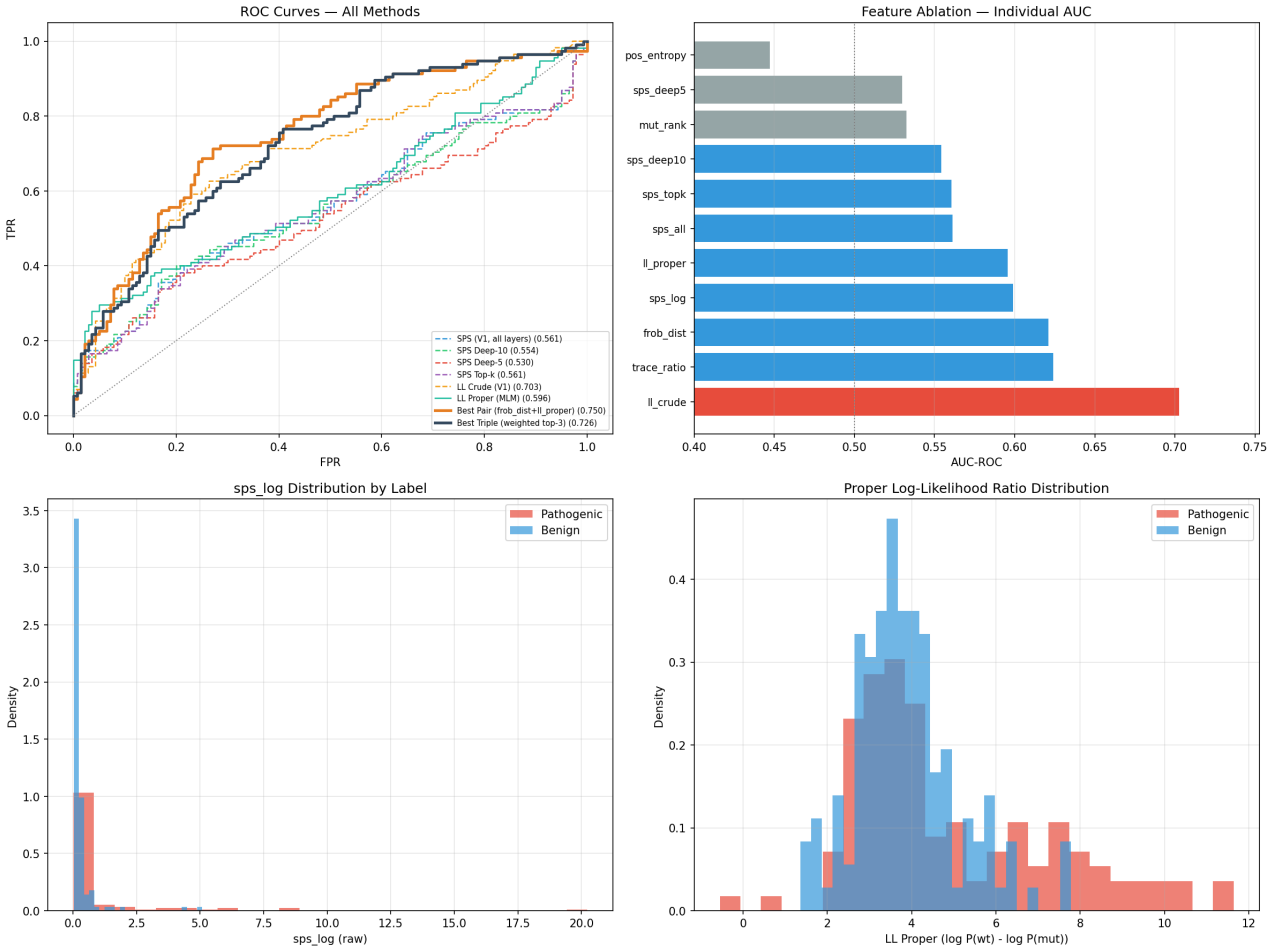

Figure 1: **SpectralBio canonical TP53 results summary (N=255).** Best Pair (FrobDist+LL Proper, orange) reaches AUC=0.7498 and remains above all individual features and the Best Triple (black, 0.7264, rounded in the figure). Matrix-level features outperform eigenvalue-only SPS variants on TP53, while the proper-LL distribution is consistent with the bounded BRCA1 secondary transfer result (AUC=0.9174). Figure linked to canonical asset `outputs/canonical/roc_tp53.png`; TP53 reproduces with Δ_{rep}=0.0.

Large-scale benchmarks for protein fitness prediction and design.
NeurIPS, 36.